# New database changes — MySQL

CableBill: CodeIgniter → Laravel migration | 11 Nov 2026 | Applied so far to the local copy `cablebill02nov` only. The live database has not been touched.

> **Only one structure change exists:** the index `idx_customer_id` on `m_ezybill_invoices` (approved). No tables, columns or views are added, renamed or dropped. Everything below is plain MySQL (5.7 / 8.x, InnoDB).

## 1. Required change APPROVED applied to local copy

Adds an index so the "latest invoice / package per customer" lookup (600,000 rows) runs in milliseconds instead of about 26 seconds. Non-breaking: the old CodeIgniter site keeps working and also gets faster.

**1a. Plain statement**

```
ALTER TABLE `m_ezybill_invoices`
  ADD INDEX `idx_customer_id` (`customer_id`);
```

**1b. Safe version (can be run more than once)** — file `database/sql/001_idx_ezybill_invoices_customer_id.sql`

```
SET @exists := (
  SELECT COUNT(1)
    FROM information_schema.statistics
   WHERE table_schema = DATABASE()
     AND table_name   = 'm_ezybill_invoices'
     AND index_name   = 'idx_customer_id'
);
SET @sql := IF(@exists = 0,
  'ALTER TABLE m_ezybill_invoices ADD INDEX idx_customer_id (customer_id)',
  'SELECT "idx_customer_id already exists"');
PREPARE stmt FROM @sql;
EXECUTE stmt;
DEALLOCATE PREPARE stmt;
```

**1c. Check that it exists**

```
SHOW INDEX FROM `m_ezybill_invoices` WHERE Key_name = 'idx_customer_id';

-- the optimiser should now use it (type = ref/range, key = idx_customer_id):
EXPLAIN SELECT customer_id, MAX(id)
          FROM m_ezybill_invoices
         WHERE customer_id IN ('1','2','3')
         GROUP BY customer_id;
```

**1d. Undo (only if ever needed)**

```
ALTER TABLE `m_ezybill_invoices` DROP INDEX `idx_customer_id`;
```

| Item | Detail |
|---|---|

| | |
|---|---|
| Table / column | `m_ezybill_invoices.customer_id` — `varchar(45)`, no index before |
| Time to run | about 3.4 seconds on the local copy (600k rows); the table is briefly busy while the index builds |
| Result seen | Subscriber list page 24 s → 1.7 s, identical output |
| Verified | Compared with database `malanadcablebill`: this index is the only structural difference between the two (tables and columns identical) |
| Go-live | **Run section 1b on the live database** during the cut-over window (after a backup, like any change) |

## 2. Optional change NOT APPROVED – NOT APPLIED

The monthly SMS counts and several reports filter `subscriberpayments` (about 990,000 rows) by `timestamp` with no index (about 3 seconds per month). Same kind of non-breaking change as section 1. Needs your approval first.

```
ALTER TABLE `subscriberpayments`
  ADD INDEX `idx_timestamp` (`timestamp`);

-- undo:
ALTER TABLE `subscriberpayments` DROP INDEX `idx_timestamp`;
```

`timestamp` is a `varchar(50)` holding milliseconds since epoch, so the index is on the text value (all values have the same length, so text order equals time order).

## 3. Cut-over only NOT APPLIED

**Password hashing.** Passwords are plain text in `m_user.password` (`varchar(150)`) and `agent.password` (`varchar(200)`). A bcrypt hash is 60 characters, so **no ALTER TABLE is needed**. The conversion itself is done by a small PHP script at cut-over (MySQL cannot create bcrypt hashes) and only after the old site is switched off, because the old site could no longer log in afterwards. Queries to run before and after:

```
-- before: how many rows are still plain text (bcrypt hashes start with $2y$)
SELECT 'm_user' AS tbl, COUNT(*) AS plain_rows FROM m_user  WHERE password NOT LIKE '$2y$%'
UNION ALL
SELECT 'agent',          COUNT(*)              FROM agent   WHERE password NOT LIKE '$2y$%';

-- the columns are wide enough for a 60-character hash:
SELECT table_name, column_name, character_maximum_length
  FROM information_schema.columns
 WHERE table_schema = DATABASE()
   AND ((table_name = 'm_user' AND column_name = 'password')
     OR (table_name = 'agent'  AND column_name = 'password'));
```

## 4. Data touched during testing (already reverted)

No data change is part of the migration. Test rows made while checking the new screens were removed. If you test on **another copy** of the database, these queries show whether anything was left behind (all should return 0):

```
SELECT COUNT(*) FROM subscriberpayments WHERE description LIKE '%ZZCONC%' OR description LIKE
'%ZZALL%';
SELECT COUNT(*) FROM expenses            WHERE remarks LIKE 'ZZE%';
SELECT COUNT(*) FROM m_customer          WHERE lco_customer_id LIKE 'ZZLCO%' OR newid LIKE
'ZZNUID%';
SELECT COUNT(*) FROM m_area              WHERE areaname LIKE 'ZZ Test%';
SELECT COUNT(*) FROM m_package           WHERE name = 'ZZ PKG';
SELECT COUNT(*) FROM gsoft_payment_types WHERE name LIKE 'ZZ PT%';
SELECT COUNT(*) FROM gsoft_billing       WHERE current_month LIKE '2099%';
-- upigateway_history had 120 rows before testing; it must still be 120 on the local copy:
SELECT COUNT(*) FROM upigateway_history;
```

## 5. What did not change

- No new, renamed or dropped tables, columns or views (58 objects: 54 tables, 4 views). No Laravel system tables (`migrations`, `sessions`, `cache`, `jobs`).
- The old code ran `ALTER TABLE m_customer ADD COLUMN c_type INT(11) DEFAULT 1` inside a web request. The new code does **not**; the column already exists (check: `SHOW COLUMNS FROM m_customer LIKE 'c_type';` — run it on live before go-live).
- Database connection settings of the new app (not stored in MySQL): strict mode off and all values returned as text, to behave like the old driver.

## 6. Go-live checklist (database)

| # | Step |
|---|---|
| 1 | Full backup of the live database |
| 2 | Run section 1b (index on `m_ezybill_invoices`) and confirm with 1c |
| 3 | Run `SHOW COLUMNS FROM m_customer LIKE 'c_type';` — it must return a row |
| 4 | Optional: section 2 if approved |
| 5 | Password hashing (section 3) only when the old site is switched off for good |
| 6 | Change the database password and the UPIGateway key (both were hard-coded in the old source) |